

ParaSight: Enabling Privacy-preserving Sensing Data Sharing via Device-to-device Utterance-based Communication

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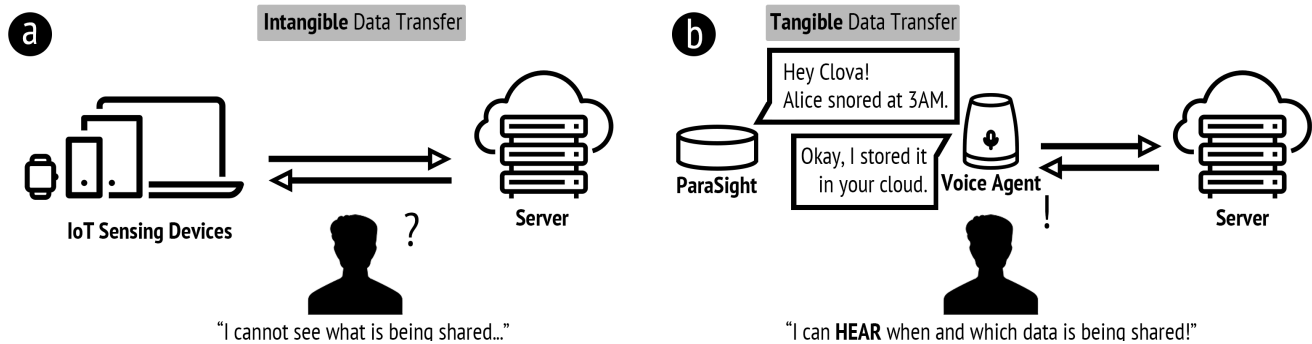


Figure 1: ParaSight enables a tangible data communication where data are transferred via human-understandable utterances. (a) End-users often fail to indicate when and which data are being communicated as existing smart home devices use invisible and inaudible data communications. (b) However, as ParaSight sends sensor data as the utterance format to a voice agent, end-users can *hear* such data transfer contents and when it happens.

ABSTRACT

Privacy-invasive sensors such as cameras and microphones in smart devices (e.g., Facebook Portal, Google Nest Hub Max, Amazon Echo Show) are now ubiquitous in a user’s private setting, which raises users’ privacy concerns of always listening and monitoring. To address such privacy concerns, we developed ParaSight, a smart speaker add-on device that communicates to users sensing data transfer. ParaSight transmits the information yet speaks out loud what is being transmitted to a smart speaker connected to a server via utterances by leveraging the smart speaker’s ability to understand human languages. Communicating sensing data as utterances brings two benefits: (i) a user can hear and understand what data, and when, are being transferred as utterances are human-understandable; (ii) raw data are processed locally and only filtered signals can be uploaded to a server. ParaSight, for example, can receive and locally process raw data of a user’s audio and video data (e.g., snoring sound, workout video), and only transmit the filtered data to a server. We demonstrate two work-in-progress

applications—snoring detection and home training applications—to show ParaSight’s use cases.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; • **Security and privacy**;

KEYWORDS

privacy-invasive sensors, privacy-preserving data sharing, conversational interaction

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1 INTRODUCTION

Activity sensing technologies in private settings often require privacy-invasive sensors (e.g., cameras, microphones) to understand and monitor a user’s behaviors. For example, health monitoring applications have started using cameras for workout activity detection and microphones for snoring detection applications [3, 6]. However, privacy concerns raised by those sensor usages have remained a long-standing problem since such technologies could put a user’s private data at risk of being accessed without their knowledge

[14, 29]. Prior work points out the drawback of the current data communication system to help explain why.

First, there is no clear way for a user to discern what sensor data and when their sensor data are uploaded and shared online, as existing device-to-server data communication technologies are often imperceptible to humans [18]. Specifically, despite claims that sensor-enabled devices are designed to avoid communicating a user's sensing data without their knowledge or consent, there is evidence to the contrary [22]. Such imperceptible data communications, in turn, cause the mismatches between one's 'perceived' and 'actual' privacy [8].

In order to address the mismatches, we developed ParaSight, a new form of a privacy-preserving sensor data communication device that enables transparent data communication to a server in a human-understandable fashion. Specifically, once ParaSight receives raw data of user activities, ParaSight locally processes them and "verbally" transmits the filtered data in a conversational format to a voice agent without sending raw data of the signal. Then, the voice agent understands the data by leveraging its natural language processing utility. For example, ParaSight receives video data of a user's workout activities, and locally processes the video to count a user's push-up repetition. Once the process is done, ParaSight transmits the push-up count data to a server by saying "Hey Alexa, the user has completed ten push-ups today." to the user's Amazon Alexa voice agent device. Then, Alexa understands and tracks the user's home training workout.

This way, ParaSight achieves the following two key benefits. First, as ParaSight "speaks" out loud what data are being transmitted to a server as utterances, a user can hear and understand what data, and when, are being communicated as utterances are human-understandable. Second, raw data are processed locally and only filtered signals can be uploaded to a server. As such, ParaSight makes the data transfer explicit and human-understandable, and negates the need of uploading raw data to a server.

As a proof-of-concept, we implemented ParaSight on Raspberry Pi 3 B+ that embeds a microphone and an RGB camera and that "speaks" locally processed data through an external speaker. We demonstrate two example applications on top of ParaSight to demonstrate its practical utility: (i) a snoring detection monitoring application that classifies a recorded sound (ii) a home training application that uses video recordings to classify an activity for workout routines.

While ParaSight could address a user's privacy concerns by allowing them to hear and understand what data are being transmitted to a server, different users may want ParaSight to communicate the different amounts and types of data being uploaded to a server based on their level of privacy concerns. Thus, we wanted to understand what amount of, and what type of, sensing data in a user's private setting the user would agree on being uploaded to a server?

To do so, as a preliminary study, we conducted semi-structured interviews with 11 participants—microphone-enabled or camera-enabled application users or non-users who considered using the applications and decided not to use them. We found that as many participants cannot ensure what kind of personal data can be accessed by application service providers, they would not want identifiable data (e.g., voice audio recording, video recording of a user's

private space) to be uploaded to their servers while they feel comfortable with de-identifiable numeric data (e.g., snoring duration, push-up count). Furthermore, some participants would want a feature that asks a user for consent to upload data to a server, which helps them understand what data, and when, are being transmitted to a server.

Lastly, based on these initial findings, we discuss the current limitations and potential development of ParaSight to improve its usability and achieve ParaSight's rich interactions.

2 BACKGROUND AND RELATED WORK

2.1 Physical Privacy Operations in Ubiquitous Computing

Intangible data communication for sensor data of smart devices often fails end-users [8]. Owing to their surreptitious data sharing online, users have trouble noticing when their data transfer of their devices' sensor data takes place. For instance, there is evidence that a user's private conversations were recorded and transferred to a manufacturer's server without knowledge [14], and such incidents, in turn, raised privacy concerns about using sensor-enabled smart devices such as smart speakers [19]. To that end, researchers have studied how to design systems that unambiguously communicate privacy states of sensor usage to end-users, which allows the end-users to indicate if, and when, sensors collect their behavior data. Ahmad et al. introduced the concept of 'Tangible Privacy', a privacy design mechanism that provides end-users with physical feedback and control for privacy states of devices [8]. For example, a webcam user covers their webcam with a physical cover [21], which enables the users to unambiguously indicate that the webcam is physically blocked. Also, prior work identified transparency of data collection as one of key factors to design smart home privacy mechanisms [27, 32, 34]. Teyssier et al. proposed an anthropomorphic webcam design that mimics human-eye behaviors as a way to enable explicit communication of webcam presence and functions to webcam users [28]. Additionally, people adopted a behavior (e.g., unplugging their smart devices) to ensure their device sensors are powered off [11, 33].

While such physical operations could provide high assurance of sensor usages and non-usages to end-users, there remain challenges on implementing physical mechanisms for other sensors including a microphone—e.g., a physical cover for a microphone cannot fully block sound unless the microphone is in a vacuum. Thus, in our work, we aim to tackle such challenges by designing a physical mechanism for data sharing instead of physical block for sensors, which helps end-users to identify what data, and when, are shared.

2.2 Auxiliary Privacy-aware Devices for Smart Devices

Smart devices have pervaded a user's daily life routines. As they use sensors that capture a user's activities and send them online (e.g., cloud), the user's privacy has become increasingly vulnerable to being exposed online without knowledge.

To that end, researchers have explored various ways to suppress such vulnerabilities by deploying an additional device around smart devices. Chen et al. demonstrate a wrist-worn wearable that

produces inaudible acoustic jamming signals to jam smart device microphones when a user wants to ensure no sensing device could eavesdrop on their private conversation [13]. Likewise, Sun et al. create a smart speaker jamming device that sits around a smart speaker and inaudibly jams its microphone when a user wants to have private conversations, which only allows a user's voice input when in use [26]. Additionally, Karmann et al. demonstrate Project Alias, which is a smart speaker add-on device that allows a user to change the smart speaker's wake-up word while jamming the smart speaker when not in use [5].

Unlike such inaudible jamming devices that thwart sensors' data collection, our focus is to investigate how to make data transfer explicit. In doing so, end-users can examine which data are uploaded online or which data could not.

3 UNDERSTANDING PRIVACY CONCERNS FOR DATA SHARING OF TARGET APPLICATIONS

Sensor data communication is invisible and inaudible to humans that users cannot clearly distinguish what data are shared online. The goal of our study was to understand what type and amount of data shared online raise privacy concerns of users of apps that use privacy-invasive sensors (e.g., microphone, camera). To do so, we chose two applications for participant recruitment: snoring detection applications that represent a microphone-based application, home training applications that represent a camera-based application. Ultimately, the privacy-preserving data type and granularity of the two applications we found are applied to the proof-of-concept implementations of ParaSight.

3.1 Method

3.1.1 Recruitment. We conducted a semi-structured interview with 11 participants physically located in Korea—application users or non-users who considered using the applications and decided not to use them.¹ We included both users and non-users of the applications to examine if both groups of participants may have different levels of privacy concerns based on the granularity of data type and amount to be accessed without their knowledge. During the recruitment, we did not explicitly inform them that the study is related to privacy to elicit their unbiased responses [9].

3.1.2 Procedure. We conducted the interviews over a video conferencing call. We asked participants two broad categories of questions: (i) their perception of how the existing apps' data are transferred and stored; (ii) the amount and type of data of the apps they would agree on being uploaded online and accessed by corporate or third-party organizations.² We designed our interview to be 30 minutes long and provided 5,000 won (approx. 4.3 USD) as compensation for participants' interview completion.

3.1.3 Data Analysis. We performed qualitative coding on the interview transcripts and used thematic analysis to identify themes in participants' perception regarding data transfer in their app usage [31]. One of the team members categorized responses from the transcripts into two groups: (i) the perception of the apps' data

transfer and storage; (ii) the granularity of data type and amount that participants would not want the application provider to access. The researcher used an open coding process on each category and iteratively updated the codebook as necessary. Then, a second member of the research team coded the interview data following the codebook. Lastly, the researchers used an axial coding process to find larger themes by identifying the connection of the codes.

3.2 Results

In general, participants felt unsure about where and how their sensor data have been processed and stored. We found that this uncertainty raises their privacy concerns about whether or not their private information could be accessed by the app provider and the third party without knowledge. While such findings echoed results studied in prior work[16] and fairly expected, our focus is to investigate what type and granularity of data users would consider as privacy-preserving for two specific applications: (i) snoring detection application; and (ii) home training application.

3.2.1 Acceptable Data Type and Amount to be Shared. Depending on the data type that participants considered to be accessed without knowledge, participants expressed different levels of privacy concerns. To be specific, most of them expressed discomfort when they think that their privacy-sensitive (identifiable) data such as voice-recorded and/or video-recorded data are uploaded and can be accessed by the app service provider while they would feel comfortable in responding to de-identifiable numeric data (e.g., numeric data including counts of exercises, the duration for snoring).

Snoring Detection Application. Most participants agreed that quantitative data such as snoring start time, snoring period, and snoring loudness are de-identifiable. With regard to audio recordings, participants showed privacy concerns particularly about the possibility of any conversational data being recorded. For example, P5 expressed concerns about the risk where the snoring detection system could misclassify sleep talking as snoring. P2 and P3 would feel comfortable if only snoring sound without any conversation sound is uploaded and shared.

Home Training Application. Similar to the responses about snoring detection apps, home training app interview participants mentioned that quantitative data including the number of repetitions of completed workout sessions, estimation of the number of calories burned would be examples of de-identifiable data. Additionally, most participants expressed a strong feeling against data containing any identifiable information including their face, their background information. Instead, they said that quantifiable measures for their body posture or silhouette would be acceptable to be shared and accessed—e.g., a user's body part position during workout compared to a reference posture.

3.2.2 Consent Interactions. Participants emphasized the need for explicit consent for data sharing online to be prevented from covert data access. Some participants also argued that if the privacy-sensitive data need to be accessed for the apps' key functionality, they would want the apps to ask the users for consent to upload data before the data are uploaded.

¹We include the demographic information in supplementary materials.

²We include the interview questions in supplementary materials.

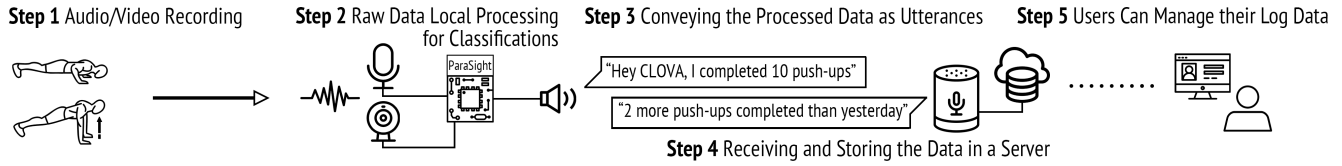


Figure 2: ParaSight receives data from sensors (e.g., microphone, camera) including microphones or cameras. Then, ParaSight locally processes the data and returns the classification output. The output is transferred as an utterance format to a voice agent such as a smart speaker. Once the voice agent receives and understands the utterance-based transferred data, the data can be uploaded and stored in a cloud server.

4 PARASIGHT

4.1 Pipeline

Our key insight of the ParaSight design is to implement a physical way to enable data communication in order to convey sensor data sharing to end-users. In doing so, end-users can indicate when their data are being shared online, and assess which data are being uploaded online. Moreover, as the limited data type and granularity we found privacy-preserving in our study are transferred, end-users can ensure only the filtered data can be communicated.

Once ParaSight’s sensors retrieve raw sound and images of a user’s activity, the data are locally processed on ParaSight. Then, the processed data are converted into the human-understandable utterance form. Lastly, ParaSight ‘speaks’ to a voice agent that is connected online by saying the utterance-form data (See Figure 2). This way, data transfer is possible without invisible data transmissions such as Bluetooth and WiFi.

4.2 Implementation

4.2.1 Hardware. Based on these design considerations, we implemented proof-of-concept ParaSight applications on a Raspberry Pi 3 Model B+ (RPi) with external hardware components: an external speaker (DFRobot FIT0502), a microphone module (ReSpeaker 2-Mic Pi HAT), and a RPi camera module (See Figure 3). Then, we placed ParaSight on a stand which allows ParaSight to be placed right above a smart speaker. This placement helps the smart speaker to be able to hear ParaSight’s utterances by placing the ParaSight’s speaker close to the smart speaker’s microphone position. For our setup, we used Clova, a smart speaker where the Korean natural language processing model operates.

4.2.2 Software. As one of the key elements of ParaSight is the ability to transfer data to other devices through utterances, we designed ParaSight to ‘speak’ to other devices as if a voice assistant user speaks to their voice assistant. To do so, we played pre-recorded audio files on the RPi including a target voice agent’s wake-up word—e.g., “Hey Alexa”, “Hey Clova”, etc. For the audio, in order to speak in a consistent intonation and tone, we used and recorded speech audios generated by the Google text-to-speech (TTS) engine for the English commands [2] and by the CLOVA TTS engine for the Korean commands [4]. Both TTS models have been widely adopted across voice assistant products [20].

While both TTS engines require the Internet connection to generate audio to synthesize voices and speak texts, we pre-recorded

all the utterances to assure that ParaSight does not use the Internet connection for TTS.

4.2.3 Classifier. As proof-of-concept applications for ParaSight, we trained a snoring detector for a microphone-based application, and a workout activity classifier for a camera-based application.

Snoring Detector. We trained our snoring detection model with the snoring dataset [17] that includes one-second-long 500 snoring audio samples and 500 non-snoring background audio samples. We featurized the dataset using Fast-Fourier Transform (FFT). To be specific, we computed the maximum power ratio, 500Hz power ratio, and spectral entropy of the audio signals using Welch’s method. We used an XGBoost classifier, which is a tree-based gradient boosting machine, for snoring detection using XGBoost library [12].

Workout Activity Classifier. We built a video dataset for the workout activity detection using a subset of the UCF101 dataset [25]. We included videos for eight workout activities, i.e., boxing punching back, jump rope, jumping jack, lunges, pull up, push up, squat, and wall push up. Total 1,001 videos were carried into our dataset, the frame rate and spatial resolution of which are 25 fps and 320×240 , respectively. We used the first 3 seconds of the videos for the activity classification.

Semantic features of the videos were extracted using ReXNet [15], which was pretrained using ImageNet dataset [24] for image classification task. The outputs of the last convolutional layer of ReXNet were obtained for individual video frames, then aggregated for each video. We trained an XGBoost classifier to distinguish eight different workout activities using the semantic features obtained from the videos.

4.3 Preliminary Evaluation

The classifiers were evaluated using a five-fold cross-validation scheme. That is, we randomly divided the data into five clusters and trained the classifier using four clusters while keeping the remained one for validation. This was repeated five times until all clusters were tested. We also optimized hyperparameters of the XGBoost classifiers based on classification performance for the validation cluster, i.e., the number of estimators, maximum tree depth, and gamma that maximize the validation accuracy are chosen for test. As a result, we obtained the average classification accuracy of 83.5% for the snoring detection and 96.3 % for the workout activity classification.

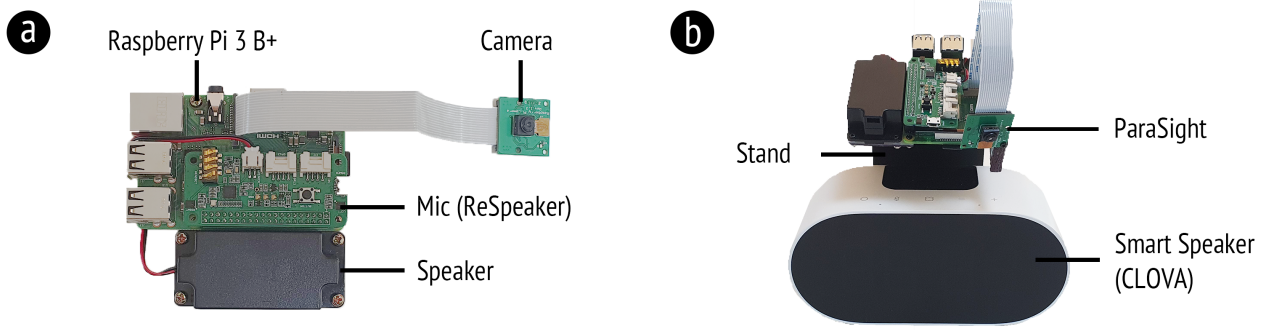


Figure 3: As the first version of the prototype, (a) we developed ParaSight with a Raspberry Pi 3 B+, a camera module, a microphone driver, and an external speaker. (b) We set up ParaSight above a smart speaker—CLOVA—by placing it on a stand that we built.

5 DISCUSSION AND FUTURE WORK

We next discuss design implications, limitations, future work, and the potential for our work to open up a broader design space of ParaSight data communication.

5.1 ParaSight Interactions

5.1.1 Usability Improvement. Usability has been considered to be an essential component to design security and privacy systems. Whitten et al. argue that the security system becomes effective when users feel sufficiently comfortable with the interface to continue using it [30]. In short, if the interactions of secure systems are not usable, end-users would be de-motivated to use the secure system and take a convenient yet insecure path to achieve their desired benefits [23]. Therefore, researchers have adopted user-centered approaches to designing security and privacy interactions [7].

For future work, we intend to study with end-users essential design components of ParaSight’s various and context-aware interactions that help the device to be more usable. For example, if ParaSight shares data about snoring during sleeping, ParaSight’s voice may surprise a user in the middle of sleeping. Therefore, instead, ParaSight can wait until the next morning and communicate the snoring data once a user consent to the data sharing to a smart speaker. Such scenarios would require ParaSight to be conversational and to understand rich context information.

5.1.2 Conversational Interactions. We could potentially leverage a smart speaker’s responding ability as well as its understanding ability to enrich the ParaSight interactions. For example, once a smart speaker listens to ParaSight’s data transfer, the smart speaker could ask a user for consent to upload the data to a server before the data upload. In addition, since the smart speaker can use data stored in the server to interact with a user, the smart speaker can provide suggestions to a user based on the data. As an example, the smart speaker can suggest the user work out their arms after realizing that the user had focused their workout on legs mostly for the past few days. This way, the ParaSight interactions could be enhanced by enabling three-way conversations: (i) ParaSight to a smart speaker; (ii) a smart speaker to a user; and (iii) a user to a smart speaker.

5.2 Potential Additional Advantages

Utterance-based data transfer can offer additional benefits. First, ParaSight could communicate with various platform voice agents and enable out-of-the-box deployment. Specifically, by adjusting a wake-up word of the ParaSight’s utterance command—“Okay, Google”, “Hey, Siri”, “Hey, CLOVA”—, ParaSight would be able to communicate processed data with a variety of voice agents such as Google Nest, Apple HomePod, and Naver CLOVA.

Second, a user can command a voice agent with other types of input including hand gestures as well as speech input. For example, when a voice assistant is playing music, a user can simply swipe their hand in front of ParaSight’s camera to play the next song. This gesture input eliminates the need of shouting to the voice assistant saying ‘Play, next!’, which helps avoid the sound interruption by the music playing during a voice command period.

5.3 Limitations of Our Implementation

5.3.1 Interaction Speed. As we ran our classification models on Raspberry Pi 3 B+, the classification runtime could cause latency for ParaSight’s real-time interactions—the current activity recognition model takes around 4 minutes to return the classification result. To address this latency, we are planning to test our model with a board designed to locally process classifications such as a tensor processing unit board [1].

5.3.2 Users versus Non-Users. In our study, we found minimal difference between users and non-users in terms of the privacy concern level according to data type to be shared online. We presume that the main reason for having little difference is that it was challenging to find and prescreen non-users who do not use such apps due to their privacy concerns. Even though we tried to prescreen non-users, no one responded that privacy concerns are the main reason for the non-adoption. This could be partially because the level of privacy concerns against microphone-enabled smart devices varies across countries. Prior work suggests that smart speaker users in Korea value the smart speakers’ utility benefits over such privacy concerns more than the U.S users [10]. Therefore, we learn that running a study with demographics over different

countries could lead to finding different implications for ParaSight interaction design.

6 CONCLUSION

We designed and implemented ParaSight that enables transferring sensor data in private settings via utterances to another device that understands natural language. ParaSight enables various interactions for users to choose what data they want to transfer and when they want to transfer the data. Through a preliminary interview study, we found that users who use microphone-based or camera-based applications showed the concerns of their private data being shared online without knowledge and consent, and discussed the type and amount of data that do not raise privacy concerns even if shared online. We envision ParaSight to serve as a privacy-preserving data communication device for exploring further privacy interactions of smart home applications.

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